

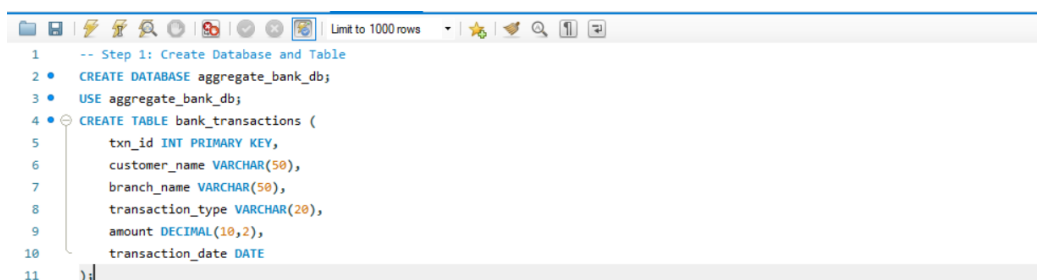
AGREEGATE FUNCTIONS

Gurrapu Nikshitha (2520030512)

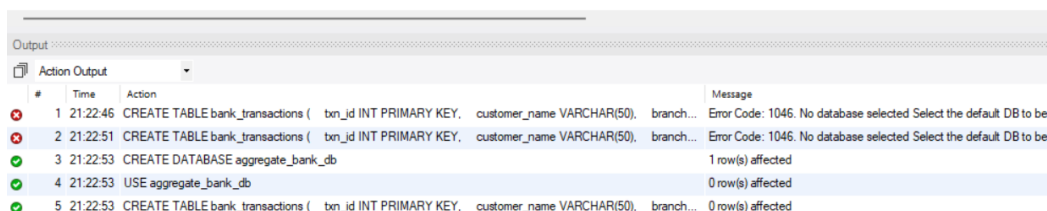
Question :

Perform SQL aggregate operations on bank transaction data using SUM(), AVG(), MAX(), MIN(), and COUNT() along with WHERE, GROUP BY, HAVING, and ORDER BY clauses to analyse transaction amounts, branches, deposits, withdrawals, and transaction counts.

Step 1 : Create Database and Table



```
1  -- Step 1: Create Database and Table
2  • CREATE DATABASE aggregate_bank_db;
3  • USE aggregate_bank_db;
4  • CREATE TABLE bank_transactions (
5      txn_id INT PRIMARY KEY,
6      customer_name VARCHAR(50),
7      branch_name VARCHAR(50),
8      transaction_type VARCHAR(20),
9      amount DECIMAL(10,2),
10     transaction_date DATE
11 )
```



#	Time	Action	Message
✗	1 21:22:46	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	Error Code: 1046. No database selected Select the default DB to be
✗	2 21:22:51	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	Error Code: 1046. No database selected Select the default DB to be
✓	3 21:22:53	CREATE DATABASE aggregate_bank_db	1 row(s) affected
✓	4 21:22:53	USE aggregate_bank_db	0 row(s) affected
✓	5 21:22:53	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	0 row(s) affected

```
12 -- Step 2: Insert Data
13 • INSERT INTO bank_transactions VALUES
14 (101,'Nikshitha','Hyderabad','Deposit',5000,'2024-01-05'),
15 (102,'Sathwika','Hyderabad','Withdrawal',2000,'2024-01-06'),
16 (103,'BhanuPriya','Vijayawada','Deposit',12000,'2024-01-08'),
17 (104,'Varsha','Vizag','Deposit',8000,'2024-01-10'),
18 (105,'Ravi','Hyderabad','Withdrawal',3500,'2024-01-11'),
19 (106,'Pranitha','Vizag','Deposit',15000,'2024-01-12'),
20 (107,'Angela','Vijayawada','Withdrawal',1000,'2024-01-13'),
21 (108,'Srija','Hyderabad','Deposit',9000,'2024-01-14'),
22 (109,'Harshika','Vizag','Withdrawal',4000,'2024-01-15'),
23 (110,'Keerthi','Vijayawada','Deposit',11000,'2024-01-16');
```

Output

#	Time	Action	Message
✗	1 21:22:46	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	Error Code: 1046. No database selected Select the default DB to b
✗	2 21:22:51	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	Error Code: 1046. No database selected Select the default DB to b
✓	3 21:22:53	CREATE DATABASE aggregate_bank_db	1 row(s) affected
✓	4 21:22:53	USE aggregate_bank_db	0 row(s) affected
✓	5 21:22:53	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	0 row(s) affected
✓	6 21:25:22	INSERT INTO bank_transactions VALUES (101,'Nikshitha','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Sathwi...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

Step 2 : AGREEGATE FUNCTIONS

```
24 -- Step 3: SUM()
25 • SELECT SUM(amount) AS Total_Amount
26 FROM bank_transactions;
```

Result Grid

Total_Amount
70500.00

Result 1 x

Output

#	Time	Action	Message
✗	2 21:22:51	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), bran...	Error Code: 1046. No database selected Select the default DB t
✓	3 21:22:53	CREATE DATABASE aggregate_bank_db	1 row(s) affected
✓	4 21:22:53	USE aggregate_bank_db	0 row(s) affected
✓	5 21:22:53	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), bran...	0 row(s) affected
✓	6 21:25:22	INSERT INTO bank_transactions VALUES (101,'Nikshitha','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Sat...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓	7 21:26:20	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned

```

27 -- Step 4: AVG()
28 • SELECT AVG(amount) AS Average_Transaction
29 FROM bank_transactions;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

Total_Amount
70500.00

Result 2 x

Output

Action Output

#	Time	Action	Message
✓ 3	21:22:53	CREATE DATABASE aggregate_bank_db	1 row(s) affected
✓ 4	21:22:53	USE aggregate_bank_db	0 row(s) affected
✓ 5	21:22:53	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), bran...	0 row(s) affected
✓ 6	21:25:22	INSERT INTO bank_transactions VALUES (101,'Nikshitha','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Sat...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓ 7	21:26:20	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 8	21:27:08	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned

```

30 -- Step 5: MAX()
31 • SELECT MAX(amount) AS Highest_Transaction
32 FROM bank_transactions;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

Average_Transaction
7050.000000

Result 3 x

Output

Action Output

#	Time	Action	Message
✓ 4	21:22:53	USE aggregate_bank_db	0 row(s) affected
✓ 5	21:22:53	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), bran...	0 row(s) affected
✓ 6	21:25:22	INSERT INTO bank_transactions VALUES (101,'Nikshitha','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Sat...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓ 7	21:26:20	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 8	21:27:08	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 9	21:28:13	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned

```

33 -- Step 6: MIN()
34 • SELECT MIN(amount) AS Lowest_Transaction
35 FROM bank_transactions;
36

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Lowest_Transaction			
1000.00			

Result 4 x

Output

Action Output

#	Time	Action	Message
5	21:22:53	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), bran...	0 row(s) affected
6	21:25:22	INSERT INTO bank_transactions VALUES (101,'Nikshitha','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Sat...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
7	21:26:20	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
8	21:27:08	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
9	21:28:13	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
10	21:28:45	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned

```

36 -- Step 7: COUNT()
37 • SELECT COUNT(*) AS Total_Transactions
38 FROM bank_transactions;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Total_Transactions			
10			

Result 6 x

Output

Action Output

#	Time	Action	Message
7	21:26:20	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
8	21:27:08	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
9	21:28:13	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
10	21:28:45	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
11	21:29:31	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
12	21:30:36	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned

Step 3 : WHERE + AGGREGATE

```

39  -- Step 8: WHERE + SUM()
40  •  SELECT SUM(amount) AS Total_Deposit
41  FROM bank_transactions
42  WHERE transaction_type = 'Deposit';

```

Result Grid | | Filter Rows: | Export: | Wrap Cell Content:

Total_Deposit
60000.00

Result 7 x

Output

Action Output

#	Time	Action	Message
✓ 8	21:27:08	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 9	21:28:13	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 10	21:28:45	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 11	21:29:31	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 12	21:30:36	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 13	21:32:07	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000	1 row(s) returned

Step 4 : GROUP BY

```

43  -- Step 9: GROUP BY + SUM()
44  •  SELECT branch_name,
45         SUM(amount) AS Total_Amount
46  FROM bank_transactions
47  GROUP BY branch_name;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
branch_name	Total_Amount		
Hyderabad	19500.00		
Vijayawada	24000.00		
Vizag	27000.00		

Result 8 ×				
Output				
Action Output				
#	Time	Action	Message	
✓ 9	21:28:13	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	
✓ 10	21:28:45	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	
✓ 11	21:29:31	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	
✓ 12	21:30:36	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	
✓ 13	21:32:07	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000	1 row(s) returned	
✓ 14	21:33:15	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name	3 row(s) returned	

Step 5 : GROUP BY + HAVING

```

48  -- Step 10: GROUP BY + HAVING
49  •  SELECT branch_name,
50         SUM(amount) AS Total_Amount
51  FROM bank_transactions
52  GROUP BY branch_name
53  HAVING SUM(amount) > 20000;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
branch_name	Total_Amount		
Vijayawada	24000.00		
Vizag	27000.00		

Result 9 x

Output

Action Output

#	Time	Action	Message
✓ 10	21:28:45	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 11	21:29:31	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 12	21:30:36	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 13	21:32:07	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIM...	1 row(s) returned
✓ 14	21:33:15	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	3 row(s) returned
✓ 15	21:34:19	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	2 row(s) returned

Step 6 : GROUP BY + ORDER BY

```

54  -- Step 11: GROUP BY + ORDER BY
55  • SELECT branch_name,
56         SUM(amount) AS Total_Amount
57  FROM bank_transactions
58  GROUP BY branch_name
59  ORDER BY Total_Amount DESC;
60
61

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	branch_name	Total_Amount
▶	Vizag	27000.00
	Vijayawada	24000.00
	Hyderabad	19500.00

Result 10 x

Output

Action Output

#	Time	Action	Message
✓ 11	21:29:31	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 12	21:30:36	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 13	21:32:07	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIM...	1 row(s) returned
✓ 14	21:33:15	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	3 row(s) returned
✓ 15	21:34:19	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	2 row(s) returned
✓ 16	21:35:45	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	3 row(s) returned

Step 7: GROUP BY + HAVING + ORDER BY


```

60 -- Step 12: GROUP BY + HAVING + ORDER BY
61 • SELECT branch_name,
62        COUNT(*) AS Total_Transactions
63 FROM bank_transactions
64 GROUP BY branch_name
65 HAVING COUNT(*) >= 3
66 ORDER BY Total_Transactions DESC;

```

	branch_name	Total_Transactions
▶	Hyderabad	4
	Vijayawada	3
	Vizag	3

Result 11 ×

Output

Action Output

#	Time	Action	Message
✓ 12	21:30:36	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 13	21:32:07	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT...	1 row(s) returned
✓ 14	21:33:15	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	3 row(s) returned
✓ 15	21:34:19	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	2 row(s) returned
✓ 16	21:35:45	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	3 row(s) returned
✓ 17	21:36:46	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...	3 row(s) returned

Step 8 : WHERE + COUNT()

```

67 -- Step 13: WHERE + COUNT()
68 • SELECT COUNT(*) AS Withdrawals
69 FROM bank_transactions
70 WHERE transaction_type = 'Withdrawal';

```

Result Grid			Filter Rows:		Export:		Wrap Cell Content:	
	Withdrawals							
	4							

Result 12 x

Output

Action Output

	#	Time	Action	Message
✓	13	21:32:07	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT ...	1 row(s) returned
✓	14	21:33:15	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	3 row(s) returned
✓	15	21:34:19	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	2 row(s) returned
✓	16	21:35:45	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	3 row(s) returned
✓	17	21:36:46	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...	3 row(s) returned
✓	18	21:37:48	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type = 'Withdrawal' LIMIT ...	1 row(s) returned

Step 9 : WHERE + AVG()

```

71  -- Step 14: WHERE + AVG()
72  • SELECT AVG(amount) AS Avg_Deposit
73  FROM bank_transactions
74  WHERE transaction_type = 'Deposit';

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Avg_Deposit			
10000.000000			

Result 13 x

Output

Action Output

#	Time	Action	Message
✓ 14	21:33:15	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	3 row(s) returned
✓ 15	21:34:19	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	2 row(s) returned
✓ 16	21:35:45	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	3 row(s) returned
✓ 17	21:36:46	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...	3 row(s) returned
✓ 18	21:37:48	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type = 'Withdrawal' LIMIT ...	1 row(s) returned
✓ 19	21:38:41	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT ...	1 row(s) returned

```

75  -- Step 15: GROUP BY + MAX()
76  • SELECT branch_name,
77         MAX(amount) AS Highest_Amount
78  FROM bank_transactions
79  GROUP BY branch_name;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
branch_name Highest_Amount			
Hyderabad 9000.00			
Vijayawada 12000.00			
Vizag 15000.00			

Result 14 x

Output

Action Output

#	Time	Action	Message
✓ 15	21:34:19	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	2 row(s) returned
✓ 16	21:35:45	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	3 row(s) returned
✓ 17	21:36:46	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...	3 row(s) returned
✓ 18	21:37:48	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type = 'Withdrawal' LIMIT ...	1 row(s) returned
✓ 19	21:38:41	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT ...	1 row(s) returned
✓ 20	22:08:34	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch...	3 row(s) returned

Step 11: GROUP BY + MIN()

```

80  -- Step 16: GROUP BY + MIN()
81  • SELECT branch_name,
82         MIN(amount) AS Lowest_Amount
83  FROM bank_transactions
84  GROUP BY branch_name;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
branch_name	Lowest_Amount		
Hyderabad	2000.00		
Vijayawada	1000.00		
Vizag	4000.00		

Result 15 x

Output

#	Time	Action	Message
✓ 17	21:36:46	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...	3 row(s) returned
✓ 18	21:37:48	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type = 'Withdrawal' LIMIT ...	1 row(s) returned
✓ 19	21:38:41	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT...	1 row(s) returned
✓ 20	22:08:34	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch...	3 row(s) returned
✗ 21	22:09:54	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions LIMIT 0, 1000	Error Code: 1140. In
✓ 22	22:10:54	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_...	3 row(s) returned

Step 12: WHERE + ORDER BY

```

85  -- Step 17: WHERE + ORDER BY
86  • SELECT *
87  FROM bank_transactions
88  WHERE amount > 8000
89  ORDER BY amount DESC;

```

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶	106	Pranitha	Vizag	Deposit	15000.00	2024-01-12
	103	BhanuPriya	Vijayawada	Deposit	12000.00	2024-01-12
	110	Keerthi	Vijayawada	Deposit	11000.00	2024-01-16
	108	Srija	Hyderabad	Deposit	9000.00	2024-01-14
•	NULL	NULL	NULL	NULL	NULL	NULL

bank_transactions 16 x

Output

#	Time	Action	Message
✓ 18	21:37:48	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type = 'Withdrawal' LIMIT ...	1 row(s) returned
✓ 19	21:38:41	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT...	1 row(s) returned
✓ 20	22:08:34	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch...	3 row(s) returned
✗ 21	22:09:54	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions LIMIT 0, 1000	Error Code: 1140. In
✓ 22	22:10:54	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_...	3 row(s) returned
✓ 23	22:13:49	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000	4 row(s) returned

Step 13: GROUP BY + COUNT()

```

90  -- Step 18: GROUP BY + COUNT()
91  •  SELECT branch_name,
92         COUNT(*) AS Customer_Count
93  FROM bank_transactions
94  GROUP BY branch_name;

```

Result Grid		
	branch_name	Customer_Count
▶	Hyderabad	4
	Vijayawada	3
	Vizag	3

Result 17 x				
Output				
Action Output				
#	Time	Action	Message	
✓	19 21:38:41	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT...	1 row(s) returned	
✓	20 22:08:34	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch...	3 row(s) returned	
✗	21 22:09:54	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions LIMIT 0, 1000	Error Code: 1140.	
✓	22 22:10:54	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_...	3 row(s) returned	
✓	23 22:13:49	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000	4 row(s) returned	
✓	24 22:14:45	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GROUP BY branch_na...	3 row(s) returned	

Step 14: HAVING + COUNT()

```

95  -- Step 19: HAVING + COUNT()
96  • SELECT branch_name,
97        COUNT(*) AS Total_Transactions
98  FROM bank_transactions
99  GROUP BY branch_name
100 HAVING COUNT(*) > 3;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	branch_name	Total_Transactions
▶	Hyderabad	4

Result 18 x

Output

Action Output

#	Time	Action	Message
✓	20 22:08:34	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch...	3 row(s) returned
✗	21 22:09:54	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions LIMIT 0, 1000	Error Code: 1140. In
✓	22 22:10:54	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_...	3 row(s) returned
✓	23 22:13:49	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000	4 row(s) returned
✓	24 22:14:45	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GROUP BY branch_na...	3 row(s) returned
✓	25 22:15:40	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...	1 row(s) returned

Step 15: GROUP BY + AVG()

```

101  -- Step 20: GROUP BY + AVG()
102  • SELECT branch_name,
103        AVG(amount) AS Avg_Amount
104  FROM bank_transactions
105  GROUP BY branch_name;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	branch_name	Avg_Amount
▶	Hyderabad	4875.000000
	Vijayawada	8000.000000
	Vizag	9000.000000

Result 19 x

Output

Action Output

#	Time	Action	Message
✗	21 22:09:54	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions LIMIT 0, 1000	Error Code: 1140. In
✓	22 22:10:54	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_...	3 row(s) returned
✓	23 22:13:49	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000	4 row(s) returned
✓	24 22:14:45	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GROUP BY branch_na...	3 row(s) returned
✓	25 22:15:40	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...	1 row(s) returned
✓	26 22:16:45	SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GROUP BY branch_na...	3 row(s) returned

Step 16: WHERE + GROUP BY + HAVING

```
106 -- Step 21: WHERE + GROUP BY + HAVING
107 • SELECT branch_name,
108        SUM(amount) AS Total_Deposit
109 FROM bank_transactions
110 WHERE transaction_type = 'Deposit'
111 GROUP BY branch_name
112 HAVING SUM(amount) > 15000;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	branch_name	Total_Deposit
▶	Vijayawada	23000.00
	Vizag	23000.00

Result 20 ×				
Output				
Action Output				
#	Time	Action	Message	
✓ 22	22:10:54	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_...	3 row(s) returned	
✓ 23	22:13:49	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000	4 row(s) returned	
✓ 24	22:14:45	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GROUP BY branch_na...	3 row(s) returned	
✓ 25	22:15:40	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...	1 row(s) returned	
✓ 26	22:16:45	SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GROUP BY branch_na...	3 row(s) returned	
✓ 27	22:17:40	SELECT branch_name, SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_t...	2 row(s) returned	

Step 17: GROUP BY + HAVING + ORDER BY


```

113      -- Step 22: GROUP BY + HAVING + ORDER BY
114      • SELECT branch_name,
115             AVG(amount) AS Avg_Amount
116      FROM bank_transactions
117      GROUP BY branch_name
118      HAVING AVG(amount) > 7000
119      ORDER BY Avg_Amount DESC;

```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	branch_name	Avg_Amount			
▶	Vizag	9000.000000			
	Vijayawada	8000.000000			

Result 21 ×				
Output				
Action Output				
#	Time	Action	Message	
✓ 23	22:13:49	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000	4 row(s) returned	
✓ 24	22:14:45	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GROUP BY branch_na...	3 row(s) returned	
✓ 25	22:15:40	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...	1 row(s) returned	
✓ 26	22:16:45	SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GROUP BY branch_na...	3 row(s) returned	
✓ 27	22:17:40	SELECT branch_name, SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_t...	2 row(s) returned	
✓ 28	22:18:22	SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GROUP BY branch_na...	2 row(s) returned	